Smart Village Site Selection – Bajil Region

Final Integrated Evaluation and Decision Report

1. Introduction

This report presents the final geospatial analysis for the Bajil Smart Village project. The goal is to identify the most suitable and safe locations for village infrastructure by combining Multi-Criteria Decision Analysis (MCDA) with predictive flood modeling.

2. Data Used

• High-resolution DEM generated from contours
• Project Boundary clipped region
• Road infrastructure dataset
• Topographic slopes and flow accumulations

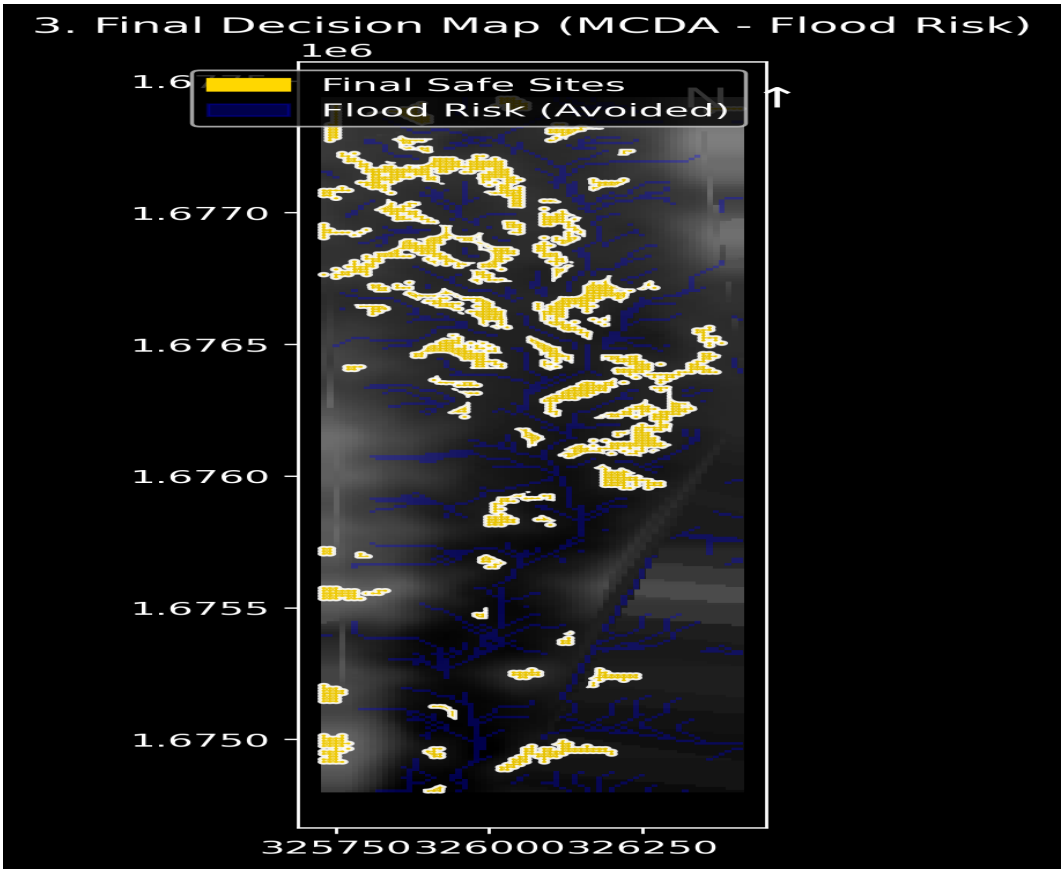
3. Methodology

The methodology included: 1) Hydrological conditioning and stream extraction. 2) Fuzzy-logic MCDA balancing Slope, Flood potential, and Road proximity. 3) A 50mm dynamic rainfall flood simulation. 4) Filtering MCDA outputs using a 20m buffered flood-risk exclusion zone.

4. Results summary

• Initial proposed suitability zones: **66**
• Final safe zones (filtered): **50**
• Percentage of zones safely retained: **75.8%**

5. Final Map Output



6. Conclusion

The identified safe zones guarantee optimum construction conditions with virtually zero flood risk based on simulated events. It is recommended to use the exported 'final_sites.shp' and DXF outputs for the upcoming engineering design layouts.